

SMARTBRIDGE / SKILL WALLET

Data Flow Diagram Architecture Mapping

Date: 2 July 2026	Team ID: —
Project Name: Caption AI : AI Powered Social Media Caption Generator	

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval/Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram for Caption AI.

EXTERNAL ENTITY USER

Initiates caption generation request by submitting text inputs and choosing target media platform profiles.

▼ Data Flow: [raw text request & platform options]

PROCESS 1 REQUEST HANDLING (VALIDATE REQUEST)

Sanitizes client form parameters, maps routing boundaries, and checks payload formatting validity.

▼ Data Flow: [verified request parameters]

DATA STORE REQUEST DATASTORE (.ENV TOKEN MAPPING)

Safely bridges verified user parameters with locked private API credentials injected from secure server configuration environments.

▼ Data Flow: [authenticated payload package]

PROCESS 2 FULFILLMENT (PROCESS REQUEST VIA GROQ CLOUD LLM)

Fires the request over remote pipeline connections to trigger sub-second natural language inference generation loops.

▼ Data Flow: [generated platform-optimized caption text]

PROCESS 3 NOTIFICATION & RELAY (NGROK TUNNEL AGENT)

Safely routes the output string back past local network firewall sandboxes directly into the presentation layout frame.

▼ **Data Flow: [final output presentation view]**

EXTERNAL ENTITY USER VIEWPORT VIEW

Displays the perfect, ready-to-copy social media caption text with custom hashtag groups directly onto the user's dashboard view.